

Seat No: _____

Enrolment No: _____



School of Computer Applications
Department of Master of Computer Applications
University Examination Summer – 2023

Subject Code: 40110201

Date: 20/02/2023

Subject Name: Data Structure

Duration: 02:30 Hours

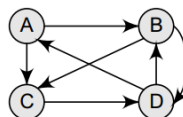
Total Marks: 70

Q1. Define following terms with suitable examples [draw suitable diagram whenever required]:

- I. Spanning Tree (2)
- II. Space complexity (2)
- III. Algorithm (2)
- IV. Level of tree (2)
- V. Balanced Factor (2)
- VI. Degree of a vertex (2)
- VII. List out any four applications of stack. (2)

Q2. Answer The Following Questions. [Attempt any four]

- I. What is Queue? Write an algorithm for inserting an element into circular queue using array. (3)
- II. Consider a two-dimensional array $\text{arr}[10][10]$ which has base address = 1000 and the number of bytes per element of the array = 2. Now, compute the address of the element $\text{arr}[8][5]$ assuming that the elements are stored in column major order. (3)
- III. Find adjacency matrix of following graph. (3)



IV. Evaluate the following postfix expression using stack: (3)

5,3,+,2,*,6,9,7,-,/-

V. In how many ways can you categorize data structures? Explain each of them with example. (3)

Q3. Answer The Following Questions. [Attempt any four]

I. Define singly linked list. Write the algorithm for the deletion at last operation on singly linked list [with suitable diagram]. (3)

II. What is algorithm efficiency? How will you express the time complexity of a given algorithm? (3)

```
int function(int n)
{
    int sum=0,i;
    for(i=1;i<=n;i++)
    {
        sum=sum+i;
    }
    return sum;
}
```

III. Convert the following expression from infix to postfix using stack: (3)

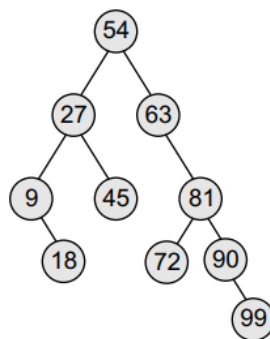
$((A+B)^C-D*(E-F/G))$

IV. Consider the following Binary Search Tree and perform following deletion operation. [use original tree in every delete operation] (3)

1.Delete 18

2.Delete 90

3 Delete 54

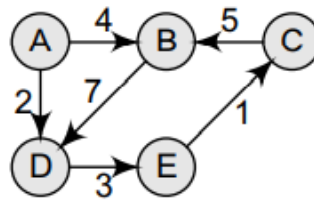


V. Define: 1). Complete binary tree 2) Isolated vertex 3) Self loop(graph) (3)

VI. Sort the following data using Radix sort: 123,765,920,482,840,721,592,530 (Show every pass). (3)

Q4. Answer The Following Questions. [Attempt any four]

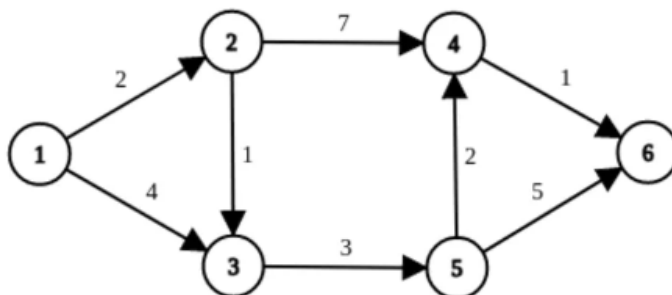
- I. Write an algorithm to insert an element at first position in circular linked list.[explain with suitable diagram] (4)
- II. Construct a binary search tree from the following data: (4)
45,23,65,89,16,28,9,36 (Show every pass).
- III. Demonstrate Quick sort on the following set of data: Take the first number as pivot [54,23,67,98,2,58,90] (Show every pass). (4)
- IV. What is hashing? Explain any one techniques for collision resolution with appropriate example. (4)
- V. Find minimum spanning tree of following graph using Prim's algorithm [Consider A as a starting vertex]. (4)



- VI. Construct an AVL tree by inserting the following elements in the given order. (4)
[50,40,30,60,55,80] By applying appropriate rotation.

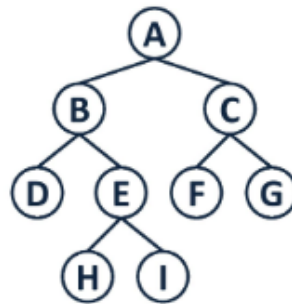
Q5. Answer The Following Questions.[Attempt any four]

- I. Find shortest path of following graph using Dijkstra's algorithm, consider 1 as source node. (4)



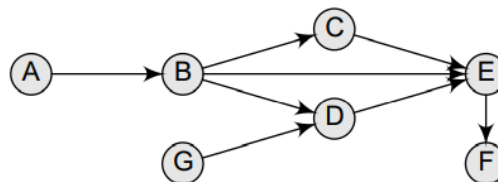
- II. Write an algorithm to insert node between any two nodes [after specified position or value] using Doubly linked list. (4)

III. List types of binary tree traversal. Find each traversal using following tree. (4)



IV. What is stack? Write an algorithm for push and pop using array. (4)

V. For what purpose the Breadth First Search (BFS) and the Depth First Search (DFS) are used? Find the DFS traversal for the following graph: (4)



VI. Arrange following elements in sorted order using insertion sort. (4)

[50,40,30,60,55,80] (Show every pass).